A local library is currently managing its books and members using a physical notebook, which has resulted in several operational problems such as misplaced books, difficulty tracking borrowed items, and inaccurate member records. To modernize its operations, the library plans to develop a database system called BookFlow. As a database developer, your task is to create the SQL foundation for this system. First, create a new database named bookflow_db. Then, design two tables: a Books table containing the columns book_id, title, isbn, and published_year, and a Members table containing the columns member_id, full_name, and email. Ensure data integrity by setting book_id and member_id as PRIMARY KEYs, applying a NOT NULL constraint to the title column so that every book must have a name, enforcing UNIQUE constraints on both isbn and email to prevent duplicate entries, and adding a CHECK constraint to ensure that the published_year is not in the future (for example, it must be less than 2027). Finally, insert three sample records into both the Books and Members tables to verify that the database structure and constraints are functioning correctly. This exercise will help you develop skills in schema definition, data integrity enforcement, and the use of SQL Data Definition Language (DDL) commands such as CREATE DATABASE, CREATE TABLE, and INSERT statements

```
mysql> CREATE DATABASE bookflow_db;
Query OK, 1 row affected (0.06 sec)

mysql> USE bookflow_db;
Database changed
mysql> CREATE TABLE Books (
    ->     book_id INT PRIMARY KEY,
    ->     title VARCHAR(255) NOT NULL,
    ->     isbn VARCHAR(20) UNIQUE,
    ->     published_year INT CHECK (published_year < 2027)
    -> );
Query OK, 0 rows affected (0.09 sec)

mysql> CREATE TABLE Members (
    ->     member_id INT PRIMARY KEY,
    ->     full_name VARCHAR(100),
    ->     email VARCHAR(100) UNIQUE
    -> );
Query OK, 0 rows affected (0.06 sec)

mysql> INSERT INTO Books (book_id, title, isbn, published_year)
    -> VALUES
    -> (1, 'The Alchemist', '9780061122415', 1988),
    -> (2, 'Atomic Habits', '9780735211292', 2018),
    -> (3, 'Clean Code', '9780132350884', 2008);
Query OK, 3 rows affected (0.03 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> ^C
mysql> INSERT INTO Members (member_id, full_name, email)
    -> VALUES
    -> (1, 'Rahul Kumar', 'rahul@gmail.com'),
    -> (2, 'Priya Sharma', 'priya@gmail.com'),
    -> (3, 'Arjun Reddy', 'arjun@gmail.com');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SHOW TABLES;
+-----------------------+
| Tables_in_bookflow_db |
+-----------------------+
| books                 |
| members               |
+-----------------------+
2 rows in set (0.03 sec)

mysql>
mysql> SELECT * FROM Books;
+---------+---------------+---------------+----------------+
| book_id | title         | isbn          | published_year |
+---------+---------------+---------------+----------------+
|       1 | The Alchemist | 9780061122415 |           1988 |
|       2 | Atomic Habits | 9780735211292 |           2018 |
|       3 | Clean Code    | 9780132350884 |           2008 |
+---------+---------------+---------------+----------------+
3 rows in set (0.00 sec)
```

```
mysql>
mysql> SELECT * FROM Members;
+-----------+--------------+-----------------+
| member_id | full_name    | email           |
+-----------+--------------+-----------------+
|         1 | Rahul Kumar  | rahul@gmail.com |
|         2 | Priya Sharma | priya@gmail.com |
|         3 | Arjun Reddy  | arjun@gmail.com |
+-----------+--------------+-----------------+
3 rows in set (0.00 sec)

mysql>
```